

The orthogonal indicator has the opposite sign Disproof of Conjecture 00000005694

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Statement and standard definition. The conjecture asserts that the symplectic type has indicator $+1$ and the orthogonal type has indicator -1 . We use the standard Frobenius–Schur indicator of a complex representation of a finite group,

$$\nu(\rho) = \frac{1}{|G|} \sum_{g \in G} \chi_\rho(g^2), \quad \chi_\rho(g) = \text{tr}(\rho(g)).$$

An irreducible representation is of orthogonal type when it admits a nondegenerate invariant symmetric complex bilinear form. Symplectic type refers to an invariant alternating form. This is the usual representation-theoretic meaning of these indicator types.

Actual orthogonal representation. Let $G = C_2 = \mathbb{Z}/2\mathbb{Z}$ and $V = \mathbb{C}$, with the trivial representation $\rho(g)z = z$. This is a genuine group representation: $\rho(gh) = \rho(g)\rho(h)$ and $\rho(e) = I$. It is irreducible, since a one-dimensional complex vector space has no proper nonzero subspace. For completeness, if a subspace S contains $a \neq 0$, then every $z \in \mathbb{C}$ equals $(z/a)a$ and therefore belongs to S .

The bilinear form $B(x, y) = xy$ is symmetric and G -invariant:

$$B(x, y) = B(y, x), \quad B(\rho(g)x, \rho(g)y) = B(x, y).$$

It is nondegenerate because $B(x, y) = 0$ for every y forces $B(x, 1) = x = 0$. It is not alternating, since $B(1, 1) = 1$. Thus this irreducible representation has orthogonal type.

Indicator computation. For each of the two elements $g \in C_2$, $\rho(g^2) = I_V$ has trace 1. The normalized character sum, with the actual group and actual representation, is consequently

$$\nu(\rho) = \frac{1}{2}(1 + 1) = 1 \neq -1.$$

This disproves the asserted value for orthogonal type and hence the conjunction in the conjecture. The fixed-point-dimension and signature clauses cannot rescue this failed clause. In fact the conventional signs are orthogonal $+1$ and symplectic -1 ; only the exhibited orthogonal counterexample is needed here.

Formalization and reference. The Lean project constructs C_2 as the multiplicative version of `ZMod 2`, uses `Representation.trivial`, obtains its character from `LinearMap.trace`, computes the finite normalized indicator sum, and proves irreducibility and all bilinear-form properties above. The final theorem combines these facts with the failure of the proposed orthogonal sign. The standard convention is documented in B. Conrad, *Frobenius–Schur indicator*, Stanford Math 210B, course handout. The counterexample is self-contained.